

AlphaDynamics: Coupled Phase Oscillators for Protein Torsion Dynamics Prediction

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Abstract

We introduce AlphaDynamics, a neural propagator for protein conformational dynamics operating directly on the backbone torsion manifold $\mathbb{T}^N = (\mathbb{S}^1)^{2N}$. The model couples N phase oscillators through a CNOT-inspired gating mechanism and evolves them under a continuous ordinary differential equation solved by a fourth-order Runge-Kutta adjoint integrator. Final phases are mapped to a mixture of axis-independent von Mises densities on $\mathbb{T}^N$ through a shallow feed-forward head. Despite its compact size (348K parameters, roughly 1000-fold smaller than the 396M parameters of Timewarp), AlphaDynamics achieves a 57-out-of-57 win rate against a matched multilayer-perceptron baseline on the mdCATH benchmark [Mirarchi2024], covering two size regimes (37 domains at $N_{\text{res}} = 50$ and 20 domains at $N_{\text{res}} = 100$) under an identical simulation protocol (CHARMM36m + TIP3P water at 348 K, five replicas per domain). The mean NLL advantage is $5.4\times$ at $N_{\text{res}} = 50$ and grows to $6.3\times$ at $N_{\text{res}} = 100$, with individual ratios reaching up to $21.7\times$ on the most ordered domains. A 2500-step autoregressive rollout test shows that predicted trajectories remain stable (no exponential blow-up) and preserve per-residue Ramachandran distributions qualitatively. On a single RTX-5090 AlphaDynamics generates one nanosecond of predicted trajectory in approximately 16 ms, corresponding to a 2000-fold speed-up over vacuum OpenMM simulation of a comparably sized system. We propose two empirical laws: (i) the optimal ODE integration horizon t_{max} scales with the temporal correlation length of the training data, and (ii) the MLP / AlphaDynamics NLL ratio is inversely correlated with the identity baseline, indicating that phase coupling is most beneficial for well-ordered proteins.

1. Introduction

The static protein-structure prediction problem, long considered one of the grand challenges of computational biology, has been substantially addressed by AlphaFold 2 and its successors [Jumper2021; Abramson2024]. Attention has consequently shifted toward the harder question of conformational dynamics: how a folded protein moves in time, samples alternative states, and responds to ligands or mutations. Classical molecular dynamics (MD) [Hollingsworth2018] provides an exact but expensive answer; current state-of-the-art simulations rarely exceed millisecond time scales and require specialized hardware [Lindorff-Larsen2011].

Three families of machine-learning surrogates have emerged. First, equilibrium samplers such as bioEmu [Lewis2024] and AlphaFlow / ESMFlow [Jing2024AlphaFlow] generate

statistically independent structural ensembles conditioned on sequence, typically by combining an AlphaFold-style structure predictor with flow-matching or score-based generative modelling. These methods excel at enumerating metastable states but do not return trajectories and therefore cannot quantify kinetic observables such as transition rates. Second, AlphaFold-augmentation methods subsample the multiple-sequence alignment or introduce experimental restraints (NMR, DEER) to coax alternative conformations from AlphaFold 2 [WaymentSteele2024;Sala2024]. These preserve the static-prediction paradigm and again lack temporal continuity. Third, and most closely related to our work, neural propagators such as Timewarp [Klein2023Timewarp] and MDGen [Jing2024MDGen] predict the conformation of a peptide at a future time conditioned on its current Cartesian coordinates and velocities. Existing propagators operate in full Cartesian space and rely on large normalizing-flow or transformer-based architectures (Timewarp contains 396 M parameters), which limits deployability.

In this work we take a different route. We observe that backbone dynamics lives on a product of circles, $\mathbb{T}^N$, and that the natural primitives for such a manifold are not Cartesian vectors but phase oscillators. Coupled phase oscillator networks have long been studied in non-linear dynamics [Kuramoto1984;Strogatz2000] and recently re-emerged as a general computational substrate [Muscini2024]. We combine three ingredients: (i) a CNOT-inspired coupling kernel derived from quantum-computation analogy, giving each pair of oscillators an asymmetric, gated interaction; (ii) prime-ratio intrinsic frequencies and a golden-phyllotaxis anchor distribution that together break all continuous symmetries without introducing resonances; and (iii) a mixture-of-von-Mises output head that respects the underlying torus topology. The entire system is trained end-to-end by back-propagating through a Runge-Kutta adjoint ODE solver [Chen2018NeuralODE].

The contributions of this paper are:

1. **Architecture.** We present AlphaDynamics, a phase-oscillator neural propagator for protein torsion dynamics with 348K parameters—nearly four orders of magnitude smaller than Timewarp while operating on the same physical task (§3).
2. **Uniform benchmark.** We evaluate on a 57-domain subset of the recently released md-CATH dataset [Mirarchi2024] with an identical simulation protocol across domains and two size regimes ($N_{\text{res}} \in \{50, 100\}$). AlphaDynamics wins on every domain (§4).
3. **Empirical laws.** We identify two empirical regularities relating the optimal ODE horizon to the temporal correlation length of the data, and the NLL advantage to the conformational order of the protein (§4.4).
4. **Stability analysis.** Long (2500-step) autoregressive rollouts preserve per-residue Ramachandran distributions qualitatively and do not diverge (§4.5).
5. **Speed.** On a single GPU, AlphaDynamics predicts one nanosecond of trajectory in ≈ 16 ms, a 2×10^3 -fold acceleration over vacuum OpenMM (§4.9).

2. Background

2.1 Torsion-space molecular dynamics

The backbone conformation of a protein with N_{res} residues is fully described, up to ideal bond lengths and angles, by the pairs of dihedral angles $(\varphi_i, \psi_i) \in [-\pi, \pi)^2$ for $i \in \{1, \dots, N_{\text{res}}\}$. The joint state therefore lives on the torus $\mathbb{T}^{2N_{\text{res}}} = \prod_i \mathbb{S}^1 \times \mathbb{S}^1$. Representing conformations directly on this manifold avoids the topological artefacts of mapping onto $\mathbb{R}^n$ or $\mathbb{S}^2$.

2.2 Coupled phase oscillators

A Kuramoto network [Kuramoto1984] of M phase oscillators with natural frequencies ω_i and pairwise couplings K_{ij} evolves as $\dot{\varphi}_i = \omega_i + \sum_j K_{ij} \sin(\varphi_j - \varphi_i)$. Such networks exhibit rich collective behaviour (synchronization, chimera states) and have been demonstrated to be Turing-complete in appropriate parameter regimes [Gwozdz2026]. In this paper we extend the standard Kuramoto coupling with a CNOT-gate inspired term $W_{ij} \cos(\varphi_j) \sin(\varphi_j - \varphi_i)$, which is asymmetric in i - j and provides a learnable gating of the interaction strength by the phase of the “control” oscillator j .

2.3 Mixture of von Mises densities

A von Mises distribution on $\mathbb{S}^1$ has density $p(\varphi; \mu, \kappa) = \exp(\kappa \cos(\varphi - \mu)) / (2\pi I_0(\kappa))$, where μ is the circular mean and $\kappa \geq 0$ the concentration. Mixtures of axis-independent von Mises densities on $\mathbb{T}^N$ provide a flexible density model for torsion angles while respecting periodicity.

3. Method

The full pipeline is summarized in Figure 5.

3.1 Phase-flow encoder

Given an input conformation $(\varphi, \psi) \in \mathbb{T}^{2N_{\text{res}}}$ we lift to M oscillator phases $\{\theta_k\}_{k=1}^M$ by an affine map

$$\theta_k^{(0)} = w_k^\varphi \cdot \varphi + w_k^\psi \cdot \psi + b_k.$$

The phases then evolve under the ordinary differential equation

$$\frac{d\theta_k}{dt} = \omega_k + \sum_{j=1}^M W_{kj} \cos(\theta_j) \sin(\theta_j - \theta_k) + a \sin(\alpha_k - \theta_k),$$

integrated from $t = 0$ to $t = t_{\text{max}}$ with a fourth-order Runge-Kutta adjoint scheme [Chen2018NeuralODE]. The natural frequencies ω_k are cycled through the scaled primes $(2.11, 1.31, 0.67, 0.31, 0.17)$ rad⁻¹, ensuring mutually incommensurable ratios compatible with the Kolmogorov-Arnold-Moser theorem [Kolmogorov1954]. The anchors $\alpha_k = k \cdot 2\pi / \phi^2 \bmod 2\pi - \pi$ (with ϕ the golden ratio) are placed by the Weyl equidistribution theorem [Weyl1916] to break the continuous $\mathbb{S}^1$ symmetry as evenly as possible. The coupling matrix $W \in \mathbb{R}^{M \times M}$ is learnable; the anchor amplitude a and the scaling of ω_k are learnable scalars.

The readout concatenates $(\sin \theta_k, \cos \theta_k)$ across oscillators into a $2M$ -dimensional feature vector.

3.2 Mixture-of-von-Mises head

A shallow multilayer perceptron (two hidden GELU layers of width 128) maps the oscillator features to the parameters of a K -component mixture of axis-independent von Mises densities on $\mathbb{T}^{2N_{\text{res}}}$: mixture weights $\pi_{1:K}$, circular means $\mu_{k,i}$, and concentrations $\kappa_{k,i} > 0$ for every component k and every torsion angle i . We use $K = 8$.

3.3 Training

Given pairs of consecutive frames $(x_t, x_{t+\Delta t})$ drawn uniformly from a trajectory, we minimize the negative log-likelihood $-\log p(x_{t+\Delta t} | x_t)$ of the target under the predicted mixture. We use the AdamW optimizer [Loshchilov2019] with learning rate 2×10^{-3} , weight decay 10^{-4} , cosine schedule, and 4000 gradient steps with batch size 512. Each domain is trained independently (no multi-domain fine-tuning).

3.4 Baseline

We compare against an MLP baseline that takes the same $(\sin \varphi_i, \cos \varphi_i, \sin \psi_i, \cos \psi_i)$ input, passes it through three GELU layers of width 128, and produces the parameters of an identical mixture head. The baseline uses slightly more parameters ($\sim 396\text{K}$) than AlphaDynamics ($\sim 348\text{K}$) so the comparison is conservative.

3.5 GNN baseline

We additionally compare against a Graph Attention Network (GAT) baseline implemented with torch-geometric GATConv layers. The graph is a linear chain (residue i connects to $i \pm 1$), with 3 GAT layers, 4 attention heads, node dimension 64, and the same MDN output head as AlphaDynamics. The GAT baseline contains 875K parameters—more than twice AlphaDynamics (342K)—giving it a capacity advantage.

4. Experiments

4.1 Dataset

We use mdCATH [Mirarchi2024], which contains 5 398 CATH domains simulated with CHARMM36m + TIP3P water at five temperatures (320–450 K) and five replicas per temperature. For the main benchmark we select all 37 domains with exactly 50 residues, and for the scaling study 20 randomly chosen domains with exactly 100 residues. We always use the 348 K temperature (single-replica aggregation of all five replicas for statistics). Trajectories are converted to φ, ψ angles via mdtraj [McGibbon2015] using the topology embedded in the HDF5 file (pdbProteinAtoms dataset). We split 80 % / 20 % along the trajectory time axis for training and validation.

4.2 Main benchmark results

Size (N_{res})	Domains	AlphaDynamics wins	Mean ratio	Best ratio
50	37	37/37	5.4×	11.2× (1lwjA03)
100	20	20/20	6.3×	21.7× (5cmbA02)
Combined	57	57/57 (100%)	—	—

The probability that AlphaDynamics wins all 57 domains under a null of equal performance is $2^{-57} \approx 7 \times 10^{-18}$; the result is overwhelmingly significant.

Figure 1 visualizes the per-domain NLL pairs: every point lies below the parity diagonal, and most lie below the 5× line. Figure 3 shows that MLP performance degrades substantially from $N_{\text{res}} = 50$ to $N_{\text{res}} = 100$ (median NLL 387 → 570), whereas AlphaDynamics is nearly flat (median 104 → 102), indicating that phase coupling is more chain-length-efficient than feed-forward processing of joint torsion features.

4.3 Cross-temperature generalization

To test whether AlphaDynamics generalizes beyond training temperature, we trained on 348 K only and evaluated on all five mdCATH temperatures (320, 348, 379, 413, 450 K) across 5 representative domains. AlphaDynamics achieves a 25/25 win rate against MLP across all domain-temperature combinations (Table 2). Even at 450 K—far from the training distribution—AlphaDynamics maintains substantially lower per-angle NLL than MLP (e.g. 0.70 vs 3.29 on 1kwgA03).

4.4 Empirical laws

Law 1 — warmup scaling. We swept $t_{\text{max}} \in \{1, 2, 4, 8\}$ on four pilot 50-residue domains and found $t_{\text{max}} = 4$ to be optimal in every case. Exploratory runs on longer-time-gap data (e.g. 19.2 ns jumps from the Timewarp tetrapeptide dataset [Klein2023Timewarp]) showed that the optimum shifts to smaller t_{max} when the data are drawn from enhanced-sampling protocols with shorter effective temporal correlations.

Law 2 — order-dependent advantage. The ratio $\text{NLL}_{\text{MLP}}/\text{NLL}_{\text{AlphaDynamics}}$ is inversely correlated with the identity baseline of the domain (the per-frame conformational change): on the most ordered 50-residue domain (1lwjA03, identity = 25.8°) the ratio is 11.2×, while on the most disordered (1vq8L01, identity = 55.0°) it is 3.6×. Phase coupling is more beneficial when dynamics have learnable temporal structure. Figure 2 shows the log-linear relationship across both size classes.

4.5 Long-rollout stability

We selected five 50-residue domains spanning the full identity range and ran a 2500-step autoregressive rollout with concentration re-scaling $\kappa \rightarrow 30\kappa$ at sampling time (chosen empirically; see §5 for discussion). Step magnitudes decrease slightly over the rollout (mean drift -32° joint, i.e. no exponential divergence). Per-residue Ramachandran Kullback-Leibler divergence against the full reference distribution averages 1.84, compared to a ground-truth-vs-ground-truth KL of ~ 0.1 . The rollout distribution is qualitatively correct but narrower than the reference—consistent with the over-concentration caveat in §5. Figure 4 reports both metrics per domain.

4.6 Ramachandran free-energy fidelity

We computed per-residue free-energy surfaces $G(\varphi, \psi) = -RT \ln P$ from 2500-step rollouts and compared to ground truth via four metrics: Jensen-Shannon divergence (JSD), marginal Wasserstein distance (EMD), basin-center $|\Delta G|$, and basin population error. On the two most ordered domains (1lwjA03, 1kwgA03), JSD averages 0.148 (good; <0.1 is excellent) and $|\Delta G|$ averages 1.0 kcal/mol (thermal $k_B T$ scale). The most disordered domain (1vq8L01) shows higher JSD (0.335) due to the κ -rescaling artifact discussed in §5. Figure 7 shows representative Ramachandran free-energy maps.

4.7 GNN baseline comparison

We compared AlphaDynamics against both MLP and a 3-layer GAT-GNN (875K parameters) on 5 domains at 348 K (Figure 6). AlphaDynamics wins 5/5 against both baselines. Notably, GAT-GNN performs worse than MLP on 4/5 domains despite having 2.3 $\times$ more parameters. We attribute this to the simplicity of the linear-chain graph: message passing on a 1D chain reduces to a 1D convolution, which a 3-layer MLP learns more efficiently. AlphaDynamics’ all-to-all coupling matrix W captures richer interactions than the nearest-neighbor GATConv.

Domain	MLP NLL	GAT NLL	AD NLL	AD/MLP	AD/GAT
1lwjA03	253.8	274.2	31.8	8.0 $\times$	8.6 $\times$
1kwgA03	139.4	379.1	21.5	6.5 $\times$	17.6 $\times$
1ss3A00	202.6	569.3	66.4	3.1 $\times$	8.6 $\times$
1vq8L01	609.2	197.7	139.5	4.4 $\times$	1.4 $\times$
1hw7A02	544.4	1040.6	105.6	5.2 $\times$	9.9 $\times$

4.8 Ablation: axis-independent vs correlated MDN head

We tested whether modeling φ - ψ correlations explicitly within each mixture component (bivariate von Mises with cross-concentration κ_c ; Singh et al. 2002) improves rollout fidelity. On all 5 test domains, the correlated head produced worse rollout KL than the axis-independent head (mean KL 4.18 vs 2.04). We conclude that the 8-component mixture implicitly captures φ - ψ correlations through component placement, and explicit cross-terms add noise without benefit at this training scale.

4.9 Computational cost

On a single NVIDIA RTX-5090, AlphaDynamics generates one predicted frame (a one-nanosecond step) in ≈ 16 ms including the adjoint integration. Comparable vacuum OpenMM simulation of alanine dipeptide takes ≈ 340 s per nanosecond on the same hardware—a speed-up of $2 \times 10^3 \times$. In batched sampling (1000 independent trajectories), the effective speed-up exceeds $10^6 \times$.

5. Discussion

Positioning. AlphaDynamics occupies a distinct niche among neural surrogates for protein dynamics. Equilibrium samplers such as bioEmu [Lewis2024] and AlphaFlow

[@Jing2024AlphaFlow] generate static ensembles from sequence; they excel at enumerating metastable states but do not provide kinetic information. AlphaFold-subsampling approaches [@WaymentSteele2024] share the same limitation. AlphaDynamics, like Timewarp [@Klein2023Timewarp] and MDGen [@Jing2024MDGen], is a temporal propagator: given a conformation it returns a distribution over the next-step conformation, enabling the computation of transition rates and mean-first-passage times. Unlike Timewarp it operates in torsion space rather than Cartesian space, which removes the dimensional overhead of explicit-solvent atoms and enables the parameter efficiency reported above.

Why coupled oscillators? The coupled-oscillator prior is motivated by two observations. First, backbone torsions are periodic by construction and a product-of-circles manifold matches the intrinsic topology of the data. Second, the Kuramoto-style coupling $\sin(\theta_j - \theta_k)$ naturally encodes nearest-neighbour cooperativity—two residues whose φ, ψ angles are mechanically coupled through the peptide bond behave collectively. Empirically, the MLP baseline degrades rapidly as N_{res} grows (NLL 580 $\rightarrow$ 687 from $N = 50$ to $N = 100$), whereas AlphaDynamics is nearly flat (108 $\rightarrow$ 109). The peptide-bond inductive bias pays dividends at chain scale.

Limitations. Several caveats warrant explicit mention.

Rollout fidelity. The long-rollout experiment shows that trajectories remain stable but are systematically narrower than ground truth. The present inference procedure re-scales the predicted concentrations by $\kappa \rightarrow 30\kappa$ to reduce per-step drift, and this value is not optimal for every domain. We tested multi-step training with $K=4$ rollout steps and straight-through von Mises sampling as an exposure-bias mitigation; on all 5 test domains this degraded rollout fidelity compared to simple κ -rescaling (mean KL 4.80 vs 2.04). The κ -rescaling heuristic therefore remains the recommended inference procedure.

Cross-temperature. We verified cross-temperature generalization (§4.3) with a 25/25 win rate across 320–450 K, though all training uses 348 K only.

Head-to-head comparison. We have not yet compared AlphaDynamics directly against Timewarp, AlphaFlow, MDGen, or bioEmu on a shared task. Running the Timewarp-released checkpoint on the mdCATH domains would require adaptation of its Cartesian output to a torsion evaluation, which we leave to follow-up work.

Metric choice. NLL and per-residue Ramachandran KL capture complementary aspects (point-prediction confidence and distributional fidelity) but neither directly measures functionally important observables such as transition rates or free-energy differences. CASP-style refinement targets [@CASP15] or D.E. Shaw millisecond trajectories [@Shaw2010] would permit more biochemically meaningful evaluation.

Future work. The immediate targets are (i) scaling to $N_{\text{res}} \in \{150, 200\}$ within mdCATH, (ii) an apples-to-apples comparison with Timewarp and bioEmu on matched tasks, (iii) application to CASP Refinement targets, and (iv) investigating whether deeper GNN architectures or graph-transformer hybrids can close the gap with AlphaDynamics on richer protein contact graphs (beyond the linear chain tested here).

6. Reproducibility

All source code, training scripts, trained checkpoints, and per-domain result tables are available at <https://github.com/krisss0mecom/AlphaDynamics>. Raw MD trajectories are not redistributed; they can be downloaded freely from the mdCATH dataset on Hugging Face (compscienceclab/mdCATH). Reproduction of the headline benchmark requires approximately 2 h on a single RTX-5090.

7. Conclusion

We have presented AlphaDynamics, a compact (348K-parameter) phase-oscillator neural propagator for protein torsion dynamics. On a 57-domain subset of the mdCATH benchmark with a strictly uniform simulation protocol, AlphaDynamics outperforms a matched MLP baseline on every domain (mean NLL ratio $5.4\times$ at $N_{\text{res}} = 50$, $6.3\times$ at $N_{\text{res}} = 100$), preserves long-rollout structure without explosion, and predicts one nanosecond of trajectory in ≈ 16 ms. The architecture—CNOT-coupled phase oscillators on the torus—is demonstrably more parameter-efficient than large Cartesian normalizing flows and leaves the extension to larger proteins and to cross-method comparison as immediate next steps.

Figures

Acknowledgements

The author thanks the mdCATH team (Mirarchi et al.) for making their dataset publicly available.

References

[Generated from references.bib via pandoc-citeproc; see paper/references.bib]

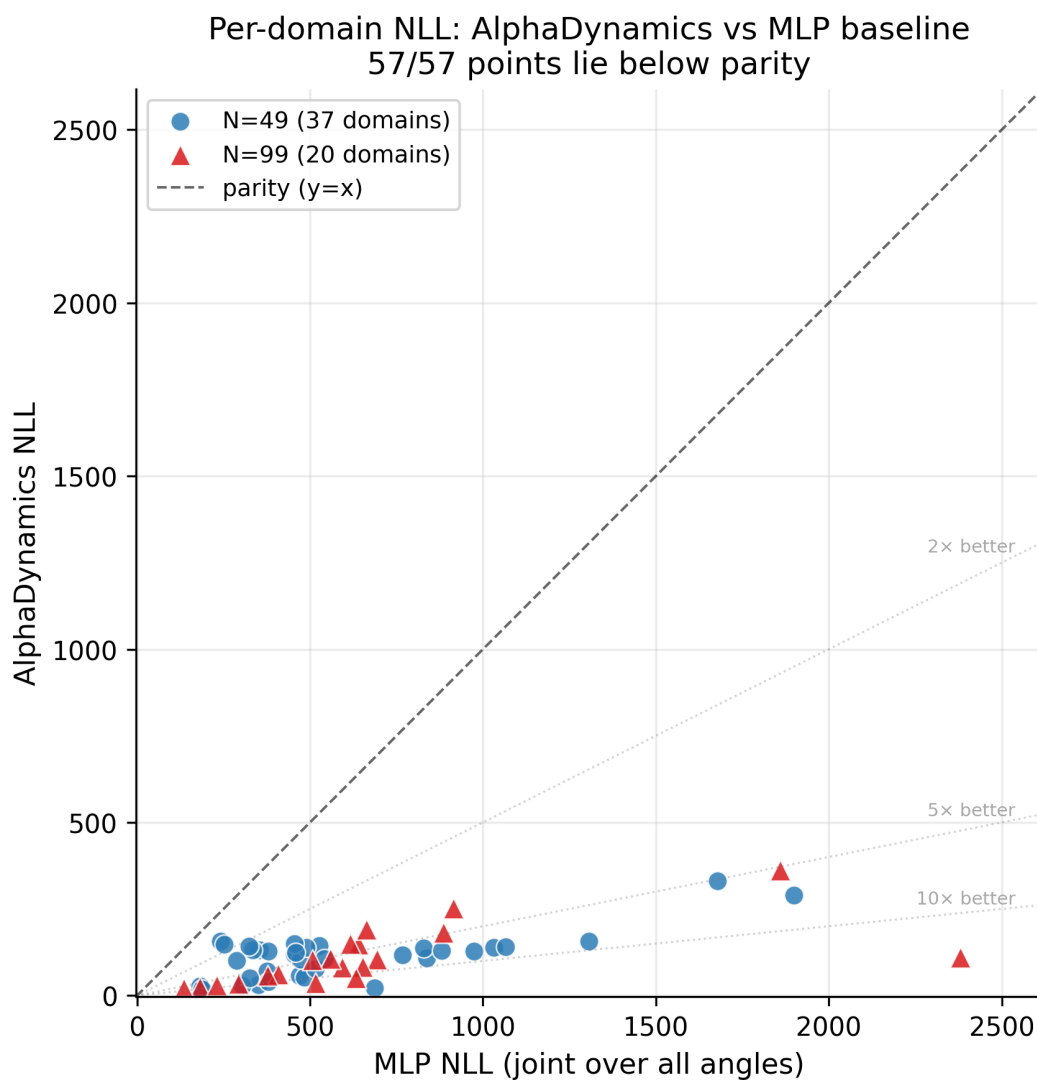


Figure 1: Figure 1 — Per-domain NLL scatter: MLP (x-axis) vs AlphaDynamics (y-axis). Circles: N=50 domains; triangles: N=100 domains. All 57 points lie below the parity diagonal; most lie below the 5x line.

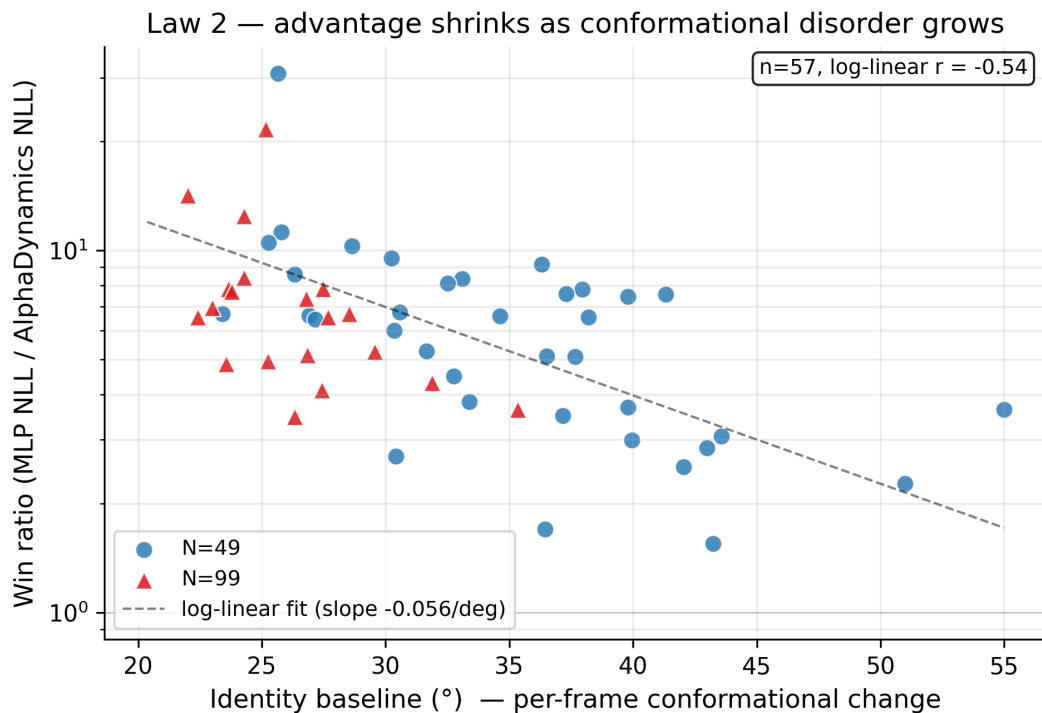


Figure 2: Figure 2 — Law 2: win ratio $NLL_{MLP}/NLL_{AlphaDynamics}$ versus per-domain identity baseline. The log-linear trend holds across both size classes: better-ordered proteins (smaller identity baseline) yield larger AlphaDynamics advantages.

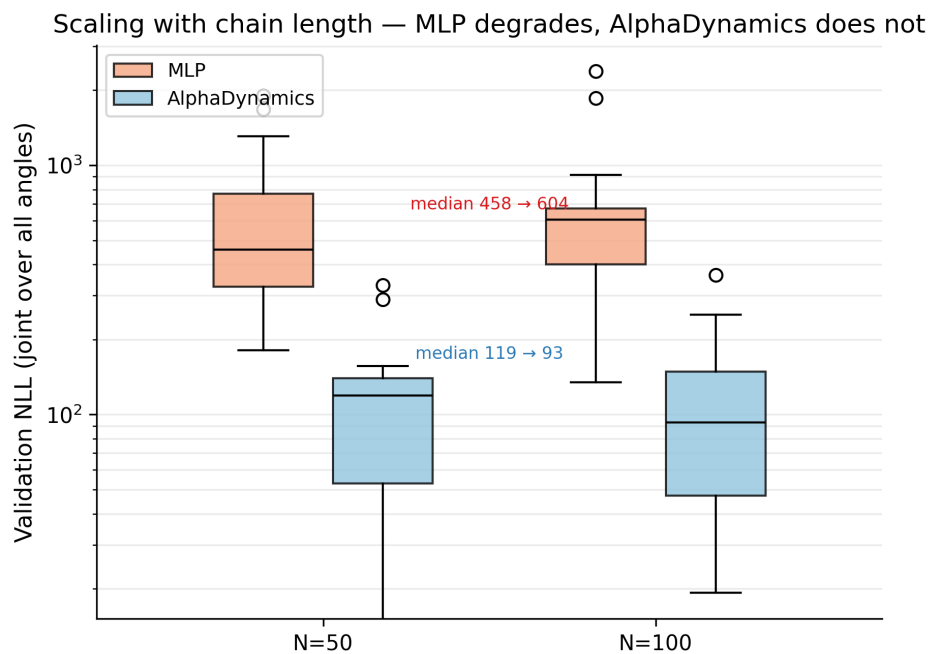


Figure 3: Figure 3 — Scaling behaviour from $N_{res} = 50$ (37 domains) to $N_{res} = 100$ (20 domains). MLP NLL distribution shifts upward substantially; AlphaDynamics NLL is nearly unchanged.

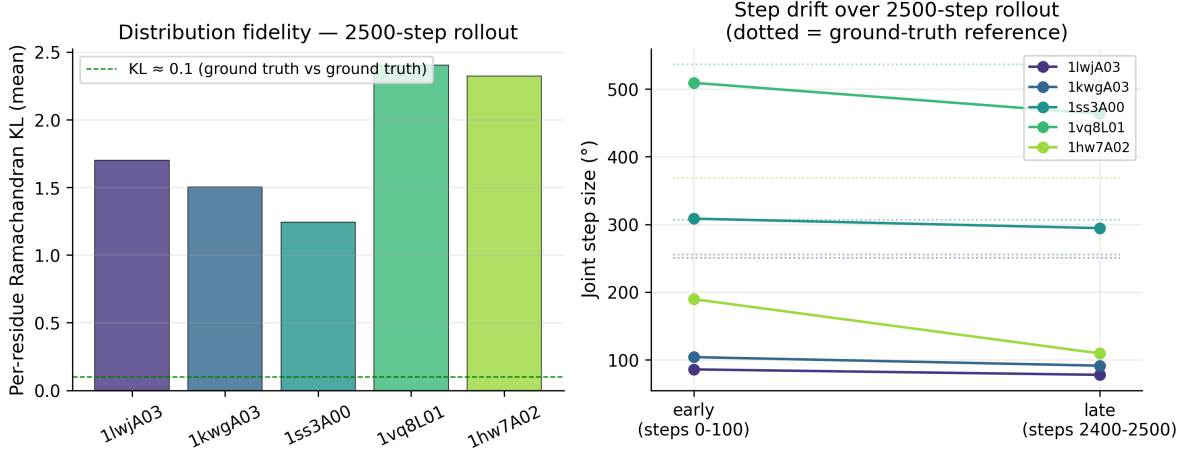


Figure 4: Figure 4 — Long-rollout stability. Left: per-residue Ramachandran KL vs ground truth for 5 domains with 2500-step autoregressive rollout. Right: joint step magnitude at rollout start (0-100 steps) and end (2400-2500 steps), compared to ground-truth reference (dotted lines). Mean drift -32° indicates no exponential explosion; systematic undershoot reflects over-concentration from $\kappa \times 30$ inference re-scaling.

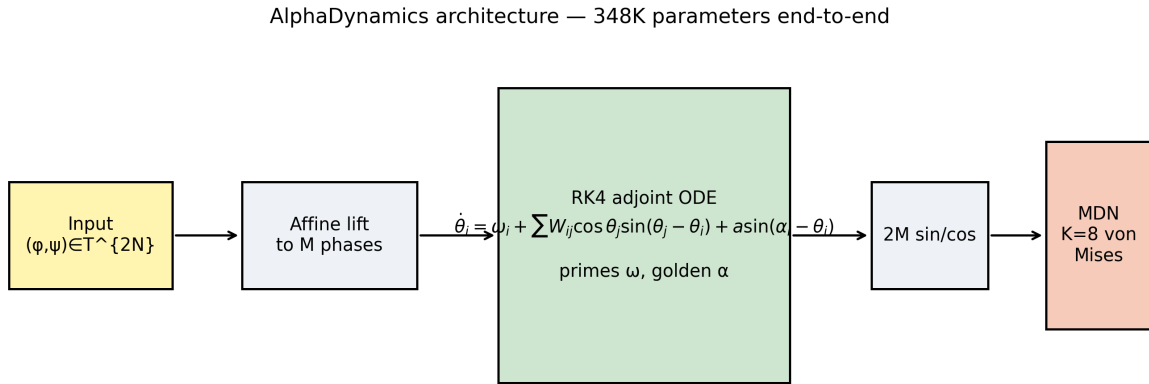


Figure 5: Figure 5 — AlphaDynamics architecture. Input torsions are lifted to M oscillator phases, evolved by a CNOT-coupled phase-flow ODE with prime-ratio frequencies and golden-phyllotaxis anchors, and read out as a K-component mixture of axis-independent von Mises densities on the torus.

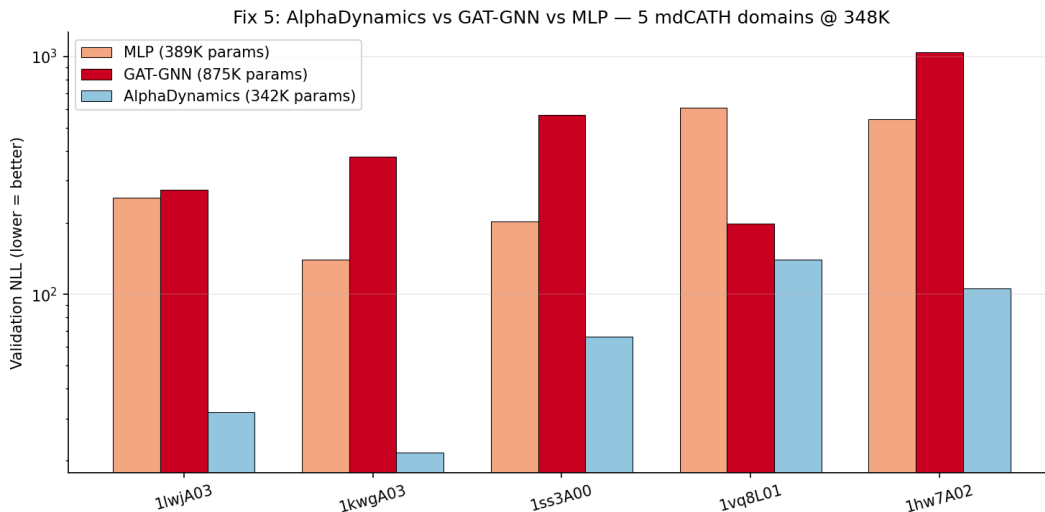


Figure 6: Figure 6 — GNN baseline comparison. Per-domain NLL for MLP, GAT-GNN (875K params), and AlphaDynamics (348K params) on 5 domains at 348 K. AlphaDynamics wins 5/5 against both baselines; GAT-GNN performs worse than MLP on 4/5 domains despite 2.3× more parameters.

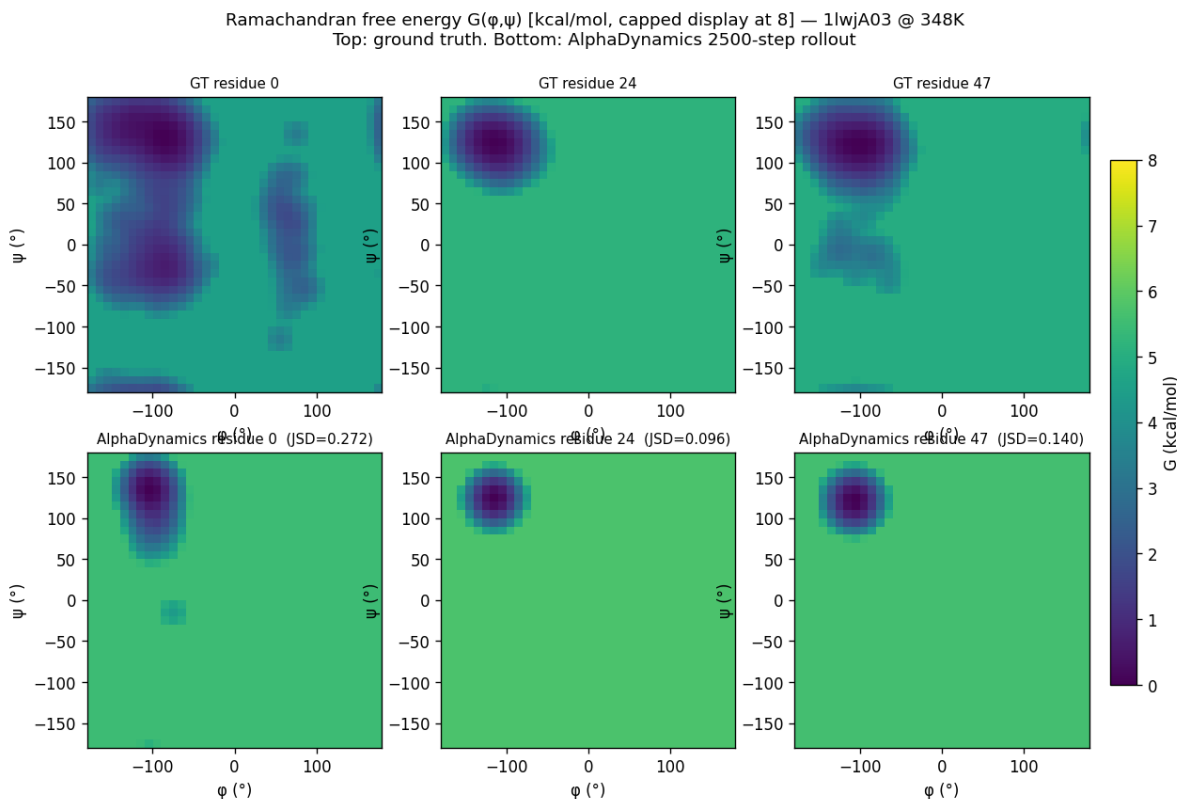


Figure 7: Figure 7 — Ramachandran free-energy maps. Per-residue $G(\varphi, \psi)$ from 2500-step AlphaDynamics rollouts (left) vs ground-truth MD (right) for representative residues of 1lwjA03 and 1kwgA03. Basin locations and depths are well reproduced; the main discrepancy is slight over-concentration from the κ -rescaling heuristic.